Chenghao (Tommy) Jiang

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Education

University of Wisconsin Madison — *Sept. 2024 – Dec. 2025 (Expected)*
MS in Electrical and Computer Engineering
- **Course:** Learning based Image Synthesis, High Performance Computing, Reinforcement Learning

University of Manchester — *Sept. 2022 – Dec. 2023*
MS in Communication and Signal Processing
- GPA: 75.5/100, **Distinction** Honor

Changchun University of Science and Technology — *Sept. 2018 – June 2022*
BEng in Optoelectronic Information Science and Engineering
- GPA: 3.86/5.00, Rank:10/221

Experience

Research Assistant — *Remote*
John Hopkins University — *Sept 2025 – Present*
- Designed a ViT-based image encoder, which can disentangle a photo into illumination (extrinsic) and scene content (intrinsics), and a decoder which can reconstruct the accurate images from the combined intrinsics and extrinsic.
- Designed a DiT-based generative pipeline which takes lighting (extrinsic) as prompt and reference scene content (intrinsics) as control, achieving high-quality vivid images.

Research Intern — *Remote*
Tera AI — *Aug 2025 – Present*
- Achieved keyframe selection and periodical global optimization on Stream3R, hugely decreasing the memory consumption of streaming inference and increasing the frame capacity.
- Designed a novel visual odometry with DINOV3 encoder, accurately matches correspondences among frames and reconstructed the camera trajectory via weighted eight point algorithm.
- Collected and processed data of photos in real traffic among different time and weather.

Projects

Privacy-Aware Sensor Data for Cooperative Perception — *Madison, WI*
Supervised by Prof. Akarsh Prabhakara — *June 2025 – July 2025*
- Explored cooperative SLAM under privacy constraints using the SHARP framework, where a contributor vehicle transmits pointmap-based novel view renderings instead of raw images to balance localization performance and privacy protection.
- Set up and evaluated the *VGGT* system for SLAM task on OPV2V dataset, comparing localization accuracy across three modes: ego-only input, SHARP-generated views, and raw multi-agent input.
- Modified the CARLA simulator within OPV2V to add a depth sensor, enabling point cloud rescaling and downstream tasks like object detection based on accurate 3D structure recovery.

RoMA-SLAM: Robust SLAM System Based on Dense Matching — *Madison, WI*
Supervised by Prof. Mohit Gupta — *Feb. 2025 – May 2025*
- Designed a **RoMA-based** SLAM pipeline inspired by the structure of *MAST3R-SLAM*, leveraging dense matching as the only input modality and achieving robust camera tracking through 3D point cloud alignment using SVD-based pose estimation.
- Constructed a keyframe-driven backend with global pose graph optimization, where loop closures and inter-frame correspondences are incorporated to jointly refine camera trajectories using dense pixel-level matches.
- Implemented post-optimization triangulation for 3D reconstruction, selecting multi-view correspondences with high matching confidence to solve projection equations and recover accurate 3D point positions.

Publications

1. Bangya Liu, ***Chenghao Jiang***, Chengpo Yan, Suman Banerjee, Akarsh Prabhakara, Privacy-Aware Sharing of Raw Spatial Sensor Data for Cooperative Perception, *HotMobile 2026* (UNDER REVIEW)